

**Smt. Chandaben Mohanbhai Patel Institute of Computer Applications,
CHARUSAT**

BCA Semester-V

CAUE302 : Python Programming

Practice Questions

OOP in Python

Assignment-3

- 1. Write a Python program to create a class 'Car' with attributes brand and color. Create an object and display its attributes.**

Sample Input / Action	Expected Output
Code Implementation	Brand: Toyota, Color: Red

- 2. Write a Python program to create a class 'Circle' with an `__init__` method that initializes the radius. Add a method `area()` to calculate and print the area.**

Sample Input / Action	Expected Output
Input: Radius = 5	Area of Circle = 78.5

- 2. Write a Python program to create a class 'Rectangle'. Include methods to calculate area and perimeter.**

Sample Input / Action	Expected Output
Input: Length = 10, Breadth = 5	Area = 50, Perimeter = 30

- 3. Write a Python program to demonstrate Single Inheritance. Create a base class 'Person' and a derived class 'Student' that inherits from 'Person'.**

Sample Input / Action	Expected Output
Code Implementation	Person Name: Krupa Student Course: BCA

- 4. Write a Python program to demonstrate Multilevel Inheritance (e.g., Vehicle -> Car -> SportsCar).**

Sample Input / Action	Expected Output
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Code Implementation	Vehicle Engine Started. Car Doors Opened. SportsCar Turbo Activated.
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5. Write a Python program to demonstrate Multiple Inheritance where a child class 'Child' inherits from both 'Father' and 'Mother' classes.

Sample Input / Action	Expected Output
Code Implementation	Father's Skill: Programming Mother's Skill: Designing Child's Skills: Programming and Designing

6. Write a Python program to create a 'BankAccount' class with deposit (), withdraw (), and display_balance () methods.

Sample Input / Action	Expected Output
Initial Balance: 1000 Deposit: 500 Withdraw: 200	Current Balance: 1300

7. Write an Employee Management System using OOP. Create a base class 'Employee' and derived classes 'Manager' and 'Developer'. Override a calculate_salary() method in each derived class.

Sample Input / Action	Expected Output
Base Salary = 50000	Manager Salary = 60000 Developer Salary = 55000

8. Create a class MovieTicket with attributes movie_name, seat_no, and price. Write methods to display the ticket details and check if a ticket is affordable based on a budget.

9. Implement the __str__ magic method in a 'Book' class to return the book title and author in a readable format when the object is printed.

Sample Input / Action	Expected Output
print(Book('Python', 'Krupa Pathak'))	Title: Python, Author: Krupa Pathak

10. Ride-Sharing Booking System (Inheritance & Overriding)

Problem Statement: Simulate a platform like Uber or Ola.

Create a base class Ride with attributes: ride_id, passenger_name, distance_km, and a base fare rate calculation method.

Create derived classes EconomyRide, PremiumRide, and OutstationRide.

Override the calculate_fare() method in each class:

EconomyRide: Base fare + ₹10 per km.

PremiumRide: Base fare + ₹18 per km + Wi-Fi luxury charge (₹50).

OutstationRide: Base fare + ₹15 per km + Driver allowance charge (₹300).

Output:

Booking Economy for Karan (15 km) -> Fare: ₹150

Booking Premium for Rohan (15 km) -> Fare: ₹320

11. Problem Statement:

You are creating an alert system for college events. Students can choose to get notifications either via Email or SMS.

Create an abstract class Notification with an abstract method send_message(user, text).

Create a concrete class EmailNotification that implements send_message() to print: "Sending Email to [user]: [text]".

Create a concrete class SMSNotification that implements send_message() to print: "Sending SMS to [user]: [text]".

Output:

Sending Email to krupa@charusat.edu.in: College will remain closed tomorrow.

Sending SMS to 9876543210: College will remain closed tomorrow.